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## Neuroendocrinology and Pituitary

### NEUROENDOCRINOLOGY AND PITUITARY CASE REPORTS

***Olfactory Neuroblastoma: The Adjunctive Role of Somatostatin Agonists After Surgery***

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**Clinical Case:** A 54 years old woman who had a right-sided nasal obstruction, rhinorrhea, and sometimes purulent discharge in 2016. In 2002 she had hyperthyroidism by Graves disease and treated for two years with remission after that. The examination of the nose revealed a dark red bleeding mass filling the area from the right nasal cavity to the nasopharynx. The biochemical and serum hormonal values were in a normal range. Urine 5 -HIIAA was 2,5 mg/24h. MRI T1 and T2-weighted images revealed a mass of 4,1 x 4,2 cm, slightly higher signal in T2 and isointense in T1, that extended into the right nasal cavity and ethmoidal sinus. A biopsy was performed, which revealed neuroendocrine characteristics. Octreotide scan with SPECT showed an intense nasal uptake. Right internal maxillary artery embolization was performed. In the next two days, the tumor was resected through a right lateral rhinotomy. Histopathological examination revealed olfactory neuroblastoma, score low grade, ki67<2%. Positive immunohistochemical staining to CD56, synaptophysin, NSE, GFAP, calretinin, S-100 and chromogranin A, and negative staining to cytokeratins (CKs) (CKAPM, CKBPM, CK8/18), CD99 and Bcl-2. The resection was almost total, with no octreotide scan uptake. The post-MRI revealed a minimal residual lesion. Lanreotide treatment was introduced after surgery; the dose was 120 mg every four weeks the first year and every 8 weeks after that. No secondary effects or biochemical alterations were observed. In 2020 the octreotide scan again revealed nasal uptake showing local recurrence. This local recurrence was treated with surgery, followed by lanreotide 120 mg every 4 weeks. No residual disease has been found afterward. **Discussion:** A limited number of cases have been reported, but none of them were treated with somatostatin agonists. As in our case, the majority of the ON is slow-progressive and non-secreting. The patients mainly complain of local symptoms such as nasal congestion. Surgery is considered to be the first-line treatment for localized disease. In the case of a close margin of the lesion or a residual tumor, the recommendation was radiotherapy or chemotherapy for more advanced disease. We confirmed that the tumor expressed somatostatin receptors. The octreotide scan has been found helpful in the diagnosis and follow-up. We treated the patient with lanreotide with no progression of the disease for three years. Local recurrence could be treated with surgery, following by lanreotide treatment, no residual disease has been found to date. The somatostatin analogs may be a useful adjuvant therapy for stable disease without evident residual disease after surgery.

## Neuroendocrinology and Pituitary

### NEUROENDOCRINOLOGY AND PITUITARY CASE REPORTS

***Pachydermoperiostosis: A Rare Masquerader of Acromegaly and Thyroid Acropachy***

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**Background:** Pachydermoperiostoasis is a rare disease of prostaglandin metabolism that presents with extensive clubbing, skin thickening, and enlargement of hands and feet, often misdiagnosed as acromegaly or thyroid acropachy.
**Clinical Case:** A 29-year-old male presented with progressive enlargement of hands and feet for the past three years, with increased sweating, seborrhea, acne, arthralgia, and photophobia. On examination, facial furrowing and severe clubbing in all fingers and toes was noticed, with palmoplantar hyperhidrosis, tenderness and widening around distal forearms, and increased heel pad thickness. He had nodulocystic acne, injected right eye with ptosis, normal pupil, and visual field. Slit-lamp examination showed extensive pannus formation, with yellow nodules, suggestive of phlyctenular conjunctivitis, secondary to a systemic inflammatory process. No prominent supraorbital ridge, prognathism, proptosis, goiter, or pretibial myxedema. No family history of a similar condition. Labs showed normal cell counts and metabolic panel. Normal IGF-1 (112 ng/mL, n 84-250 ng/ml), TSH (3.82 mIU/L, n 0.27-4.2 mIU/L), fT4 (1.19 ng/dL, n 0.93-1.7 ng/dL), Testosterone (354 ng/dL, n 249-836 ng/dL) and Prolactin (15.2 ng/mL, n 2.1-17.7 ng/mL). Inflammatory markers were elevated with ESR of 54 mm/hr (n 0-15 mm/hr) and CRP of 12.45 mg/L (n 1-4 mg/L). ANA, RF, anti-CCP, and HLA-B27 were negative. C3 and C4 levels were normal as well. Interferon-gamma release assay was negative. Imaging showed a normal chest X-ray. Brain MRI showed normal pituitary gland and increased scalp skin thickness with cutis verticis gyrata "undulating skin sign." X-ray of hands and feet showed diffuse periosteal reaction. A diagnosis of pachydermoperiostosis (PDP) was made based

on clinical criteria (thickened skin, periosteal reaction, and finger clubbing). PDP is a genetic disorder of prostaglandin metabolism with variable expression and incomplete penetrance. It manifests after puberty and affects males more than females with a 7:1 ratio. PDP has been recently shown to go through an inflammatory phase, in which patients may benefit from immunosuppressants or NSAIDs. PDP should be distinguished from hypertrophic pulmonary osteoarthropathy, thyroid acropachy, and acromegaly. Our patient had a normal chest radiograph and normal hormone levels (TSH, T4, and IGF-1). The patient received topical steroids and antibiotics with artificial tears for his eyes, which resulted in a good response, and was referred to a rheumatology clinic for further management. **Clinical Lesson:** PDP can be mistaken for acromegaly or thyroid acropachy and should be considered in any patient with acromegaloid features with normal IGF-1 and no evident pituitary pathology. Awareness of this condition can help in reaching the diagnosis promptly.

## Neuroendocrinology and Pituitary

### NEUROENDOCRINOLOGY AND PITUITARY CASE REPORTS

***Panhypopituitarism Presenting as Progressive Visual Acuity Deterioration Secondary to Suprasellar Germinoma***

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**Background:** Overall the incidence of germ cell tumors worldwide is 0.1% with 60% comprising of germinomas. These may present in the suprasellar region a third of the time and thus impart variability in its presenting characteristics. **Clinical Case:** A 27-year male presented with progressive blurring of vision with incongruent visual acuity and refraction grade with optic disc pallor assessed with normal angiography. Patient also presented with increasing docility, sluggishness and more withdrawn behavior paired with polyuria, polydipsia and cold intolerance. Progressive worsening of vision led to cranial CT revealing an enhancing hyper density in the right caudate head later elucidated via MRI with several interspersed non-enhancing foci in the right caudate nucleus, both internal capsules with associated mild mass effect. The largest component located in the right capsular region with measurement of 21 x 21 x 16 mm with enhancing lesions in the hypothalamus and pituitary stalk. Endocrine evaluation revealed low serum cortisol at 91.03nmol/L (NV 138-685 nmol/L), responsive to ACTH stimulation test (65.94nmol/L to 387nmol/L), high prolactin at 856.9 mIU/L (42.4-296.8mIU/L) and low FT4 8.62pmol/l (9.01-19.05 pmol/L) despite normal TSH 1.528uIU/ml (0.35-4.94 uIU/mL). Optic nerve atrophy was now attributed to compressive intracranial mass. Secondary adrenal insufficiency was managed with hydrocortisone 50mgIV every 8 hours, and central hypothyroidism was managed with levothyroxine 75mcgtab 1tab once a day. Increased urinary output assessed as diabetes insipidus was given desmopressin 60mcg/tab ½ tab 2x a day at this time. Stereotactic biopsy of the brain revealed a germinoma of the right caudate lobe. Given the nature of the intracranial mass with high sensitivity to chemotherapeutic and radiotherapy, a multidisciplinary approach to treatment was taken with radiotherapy, hormonal and steroid replacement. Patient underwent external beam radiation therapy of the brain and spine for a total of 51 treatments allowing for improvement of visual acuity to counting, with polyuria less than 2-3 diaper changes per day. **Conclusion:** Optic disc pallor is a clinical sign that indicates optic nerve atrophy reflective of the optic tract. This may be an important clinical sign to increase clinical suspicion for intracranial mass lesion especially with correlation to manifestation of pituitary hormone deficiencies.

## Neuroendocrinology and Pituitary

### NEUROENDOCRINOLOGY AND PITUITARY CASE REPORTS

***Paraganglioma***

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**Introduction**: Paragangliomas are rare neuroendocrine tumors arising from extra-adrenal medullary neural crest derivatives. The terms pheochromocytoma and paraganglioma are often used interchangeably because morphologically and functionally these entities are almost the same. However, paragangliomas that arise in the adrenal medulla are called pheochromocytomas and those outside the adrenal gland are called paragangliomas. Paragangliomas are often discovered incidentally during imaging studies performed for other reasons. We report a case of a patient who had incidental finding of retroperitoneal paraganglioma on imaging done for evaluation of thigh and gluteal cellulitis.
**Clinical Case:** A 24-year-old female presented with chief complaints of episodic headaches, multiple skin abscesses, fever and malaise. She has a history of Type 1DM and had been non-compliant on insulin. Vitals on presentation was significant for tachycardia with heart rate of 124, blood pressure of 119/81, respiratory rate of 16. Initial labs were consistent with DKA. CT abdomen and pelvis done for further evaluation of gluteal and thigh abscess showed incidental finding of a 3.4 cm retroperitoneal/para-aortic well-circumscribed rim-enhancing mass with central hypoenhancing component, suspicious for neoplasm such as paraganglioma. Biochemical testing was performed. 24-hour urine catecholamine levels obtained showed elevated urine norepinephrine level of 1008µg/day (reference range 15-100µg/day), urine dopamine 410µg/day (reference range 65- 400µg/day), urine epinephrine less than 2µg/day (reference range 0-20µg/day). Total urine catecholamines was elevated 1008µg/day (reference range 15-100µg/day). These results confirmed diagnosis of paraganglioma. Treatment options were discussed with the patient including surgery for removal of paraganglioma which she has currently declined.
**Conclusion:** Diagnosis of a paraganglioma can usually be made using biochemical and radiographic testing. All patients with paraganglioma should be tested for hypersecretion of catecholamines in a 24-hour urine or serum collection, even if they do not present with a clinical picture of catecholamine hypersecretion. Importantly these